

Case: BioTech Innovations

Exam-style case analysis · Workshop material

BioTech Innovations (BTI) is a Boston biotechnology firm. Its flagship product, NanoSense, is a biosensor that detects early-stage cancers through a minimally invasive test at an affordable price. BTI is now in the final stages of developing NanoSense Pro, a version for critical care that monitors a patient continuously, flags sudden changes and predicts a crisis before it happens, sending live data into hospital systems. The company is seeking regulatory approval to sell NanoSense Pro to hospitals worldwide. BTI's lead over its competitors rests on four things: proprietary algorithms that analyse biological data, unique processes for manufacturing microscopic sensors, machine-learning models that become more accurate over time, and the specialised knowledge of its scientists and engineers.

BTI has invested heavily in digital transformation. Its laboratory systems, corporate IT and automated manufacturing lines are joined on a single high-speed campus network so that data can be analysed across the company in real time. BTI has no dedicated cybersecurity department. Security sits inside the Technology Operations team (BTI-Tech), led by the Chief Technology Officer, Dr Emily Chen, who reports to the CEO. The Chief Information Security Officer, Olivia Martinez, reports to Dr Chen. She maintains the security policies, runs periodic risk assessments, manages compliance with medical privacy regulations and organises a basic annual awareness course for staff.

BTI-Tech's first priority is keeping systems fast and available while the company grows, and its staff are stretched thin. BTI's security strategy is built around the network perimeter: next-generation firewalls backed by an intrusion detection and prevention system, which deal with three or four attempted breaches every month. Researchers and executives increasingly use personal devices and cloud services for their work. Dr Chen suspects that security was never fully built into the new systems introduced during the transformation.

Over the past year, BTI has dealt with four incidents. Each was handled separately, as it arose.

Four months ago, at a medical conference in Singapore, laptops belonging to several of BTI's lead researchers were stolen from their hotel rooms. The laptops held clinical trial data, proprietary algorithms and early designs for the next generation of sensors.

Eight months ago, computing nodes in BTI's research data centre crashed and rebooted repeatedly over the course of a week. These systems run BTI's drug-discovery research, including its molecular modelling algorithms and large sets of genetic data. BTI-Tech found no cause and put it down to faulty hardware. As a precaution, the team sent forensic copies of the systems to an outside security firm. Three months later, the firm reported that it had found sophisticated malware designed to target BTI's AI-driven research processes. The firm could not say what the malware did without access to BTI's research environment. The systems had been running normally for some time, so management refused the request.

Three months ago, the personal assistant to BTI's Chief Financial Officer paid a supplier \$275,000 for a new piece of laboratory equipment. The bank details on the invoice had been altered, and the money went to an offshore account that cannot be traced.

Last month, a hospital in Eastern Europe taking part in the NanoSense Pro clinical trial reported a serious incident. During a night shift, every NanoSense Pro device on the ward began sending false critical alerts, and staff rushed to treat patients who were stable. At the same time, the hospital's electronic health records system went down. When it was restored, much of the patient data the devices had collected during the trial had been deleted. BTI's response team flew to the hospital. On the hospital's network, and on the devices themselves, they found the same malware that had been found in BTI's research systems.

BTI's board has now ordered a full review of the company's information security strategy before NanoSense Pro goes to market. The board wants to know whether the current strategy is adequate, given what now looks like a capable adversary targeting the company's most valuable assets and, possibly, its patients.